

Spain

National Commission on Markets and Competition

I. Introduction: Why Bid-Rigging Detection Matters

Public procurement represents one of the largest and most strategically significant markets worldwide, accounting for over 14% of the European Union's GDP.¹⁶ However, it is also particularly vulnerable to collusion—most notably, bid-rigging—which undermines competition, inflates costs, and erodes trust in public institutions. The consequences are profound: higher prices, lower quality for taxpayers, and reduced innovation in public services.

Bid-rigging remains one of the most complex and damaging forms of anticompetitive behavior. It involves companies that, instead of competing independently, secretly coordinate their actions to manipulate the outcome of public tenders. This behavior is highly adaptive and difficult to detect, especially as collusive strategies evolve alongside technological and market changes.

In Spain, addressing bid-rigging has become a top priority for the National Commission on Markets and Competition (CNMC). The CNMC's Economic Intelligence Unit (EIU) has embraced artificial intelligence and data science as critical tools in this fight. By leveraging advanced analytics, machine learning, and network analysis, the CNMC is not only increasing the efficiency and reach of its investigations but also setting a new standard for proactive and data-based antitrust enforcement.

Critically, while advanced computational methods significantly enhance detection capabilities, human oversight and decision-making remain indispensable at every stage, ensuring accountability and fairness.

This contribution outlines the CNMC's current approach—a multi-layered system for detecting bid-rigging—and shares organizational lessons and forward-looking insights that may be of value to competition authorities and policymakers globally.

¹⁶ Organisation for Economic Co-operation and Development, Promoting the Uptake of Strategic Public Procurement in European Union-Funded Projects (2024-2026), <https://www.oecd.org/en/about/programmes/dg-regio/promoting-the-uptake-of-strategic-public-procurement-in-european-union-funded-projects-2024-2026.html>.

II. Present: A Multi-Layered System for Bid-Rigging Detection

The CNMC's approach to bid-rigging detection is anchored in a multi-layered system that combines advanced machine learning, network analytics, and cutting-edge search technologies. This integrated strategy enables the EIU to move beyond traditional, manual methods—allowing for early warning, deeper insights, and more efficient use of investigative resources.

A. BRAVA as the Core

At the centre of the system stands BRAVA (Bid-Rigging Algorithm for Vigilance in Antitrust), a supervised machine learning tool designed to flag potentially collusive tenders. BRAVA is trained using real, previously investigated and sanctioned cases, enabling it to recognize complex patterns of suspicious behavior that might otherwise go unnoticed. Importantly, BRAVA operates as a screening and prioritization tool: it does not replace human judgment but empowers analysts to focus on high-risk cases and allocate resources more strategically.

BRAVA draws on both structured and unstructured data from the Spanish Public Procurement Platform¹⁷ and the Commercial Registrars of Spain.¹⁸ It evaluates a broad set of indicators—such as the frequency with which firms bid together, unusual price similarities, or the presence of unexpected joint ventures. The system is fully explainable: using methods like LIME¹⁹ and SHAP,²⁰ every flagged case is accompanied by an explanation, making it easier for analysts (and ultimately courts) to understand the underlying rationale.

B. Graph Analytics

To complement BRAVA, the CNMC uses graph databases and network analysis to map and explore the intricate web of relationships between firms, bids, and individuals. This technology enables the detection of communities of companies that repeatedly bid together, or key intermediaries who might facilitate coordination.

¹⁷ Ministerio de Hacienda y Administraciones Públicas, Plataforma de Contratación del Sector Público, [Contratación del Estado.es](https://contrataciondelestado.es/wps/portal/plataforma), <https://contrataciondelestado.es/wps/portal/plataforma>.

¹⁸ Colegio de Registradores de la Propiedad, Mercantiles y de Bienes Muebles de España, Registradores de España, [Registradores.org](https://www.registradores.org/), <https://www.registradores.org/>.

¹⁹ LIME: Local Interpretable Model-agnostic Explanations, i.e. a method to explain individual predictions of machine learning models.

²⁰ SHAP: SHapley Additive exPlanations, i.e., a unified approach to explain output of machine learning models.

Graph algorithms from data science libraries allow analysts to uncover hidden structures and atypical patterns—going far beyond what can be seen in spreadsheets or standard databases.

By visualizing and quantifying these relationships, the CNMC can more effectively identify and contextualize clusters of collusive behavior, especially in cases where direct evidence may be lacking.

C. Advanced Search and NLP

A third pillar of the system is the use of advanced search and NLP:²¹ The gUIE.me platform. This technology allows investigators to quickly search and analyse vast volumes of both structured and unstructured data—including public tender documents and technical reports.

The integration of semantic search (using embeddings) enables the identification of similar contracts and market segments. This not only accelerates the process of finding relevant cases but also underpins market segmentation efforts by criteria such as contract type (CPV), geographic location and contract value.

III. Organizational and operational lessons learned

The transition from traditional case-by-case investigation to a data-driven, technology-enabled approach has required deep organizational and cultural change within the CNMC. Establishing the EIU meant bringing together professionals from diverse backgrounds—legal, economic, and technical—and creating a shared language and mission.

A. Building a Multidisciplinary Team

Success in deploying advanced analytics has depended on fostering close collaboration among experts with different perspectives. Lawyers, economists, data scientists, and IT specialists must work side by side to ensure that new tools meet legal standards, reflect economic realities, and are operationally robust. Investing in training and continuous learning has been crucial to overcome initial resistance and build trust in technology-driven methods.

B. Technical and Practical Challenges

²¹ NLP: Natural Language Processing, i.e. the field of AI focused on understanding and generating human language.

Integrating data from multiple sources—public procurement, company registries, and document repositories—posed significant technical and legal challenges. Data quality, consistency, and privacy safeguards all require constant attention. Likewise, model validation, auditability, and explainability are not just technical needs: they are fundamental to ensuring that screening tools are both fair and useful for analysts and, ultimately, for enforcement actions. Particular care is given to ensure that the algorithms remain unbiased and equitable.

A major lesson has been that even the most sophisticated algorithms are only as effective as the workflows and human oversight surrounding them. Early detection systems must prioritize interpretability, and ease of use for practitioners who may not have a technical background. Visual dashboards and natural language explanations have proven invaluable in bridging the gap between complex models and everyday investigative practice.

C. Organizational Impact

The adoption of these technologies has transformed the way the CNMC identifies, prioritizes, and investigates possible collusion. Rather than relying solely on external complaints or random case selection, the CNMC now proactively screens large volumes of data, allowing for more strategic allocation of resources and faster, more data-based interventions. At the same time, the importance of human judgment remains central: technology amplifies but does not replace the expertise of case handlers and investigators.

IV. Looking Ahead: Next Steps and Future Directions

The journey towards more effective and transparent bid-rigging detection is ongoing. While the CNMC's current system represents a major leap forward, several promising directions are now being explored to further enhance capabilities and meet new challenges.

A. Expanding Explainability and Human-Centric AI

One immediate goal is to deepen the use of explainable artificial intelligence (AI). The integration of generative AI for producing natural language explanations—already in use for interpreting complex models such as LIME and SHAP—is being expanded. This will make it even easier for analysts and decision-makers to

understand the rationale behind flagged cases, reducing barriers for non-technical users and ensuring transparency for courts.

B. Towards Next-Generation Detection: Modular, Multi-Agent Systems

Looking further ahead, the CNMC's modular, agent-based architecture already leverages a suite of advanced analytical modules—including Graph Neural Networks (GNN), Convolutional Neural Networks (CNN), zero-shot learning, and association rule mining—to detect collusion from multiple perspectives. Each specialized module contributes its own 'opinion' or detection signal, which are then integrated and reasoned upon, producing an explainable and robust assessment of collusive risk.

While many of these technologies are still at the research or pilot stage, they represent a clear roadmap for the future: one where the system is more adaptable, and capable of detecting both known and novel forms of collusive behavior.

In this evolving landscape, the guiding principle remains clear: new technologies must serve both the effectiveness and the legitimacy of competition enforcement, always placing human expertise and public trust at the centre.

V. Conclusion

The experience of the CNMC demonstrates that the fight against bid-rigging in public procurement is as much about institutional transformation as it is about technological innovation. Developing a robust, multi-layered detection system has required not only new tools, but also new skills, workflows, and mindsets—anchored in the values of explainability, and strategic foresight.

While AI and data science have amplified the agency's capacity to detect and prioritize collusion, they have also highlighted the ongoing importance of human expertise and judgment. The most effective solutions arise when technology and people work together, ensuring that enforcement actions are not only efficient and proactive, but also fair, understandable, and trusted by society.